

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2013-14

Paper No. MTB-609: Special Theory of Relativity-II

Time: 3 Hours

Full Marks: 70 (Symbols have their usual meanings)

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) A proton of rest mass 1.67×10^{-24} g is moving with velocity $0.995c$. Calculate its energy and momentum. [4]

- (b) Show that the momentum p of a particle of rest mass m_0 and kinetic energy K is given by:

$$p = \sqrt{\frac{K^2}{c^2} + 2m_0K} \quad [4]$$

- (c) Obtain the transformation equations for $\vec{H}$ (the magnetic field). [4]

- (d) Prove that $\vec{E} \cdot \vec{H}$ is Lorentz invariant. [4]

- (e) Find F_{12} and F_{13} in terms of $\vec{E}$ and $\vec{H}$. [4]

- (f) State the expression for the Lorentz force density. [2]

Question 2.

- (a) Establish the transformation formula for relativistic mass and hence prove that $F = \frac{dp}{dt}$. [6]

- (b) Two particles with proper masses m_1, m_2 move along the x -axis with velocities u_1, u_2 . They collide and coalesce. Using conservation of relativistic momentum and energy, prove that the proper mass m_3 and velocity u_3 of the resulting particle satisfy:

$$m_3 = \sqrt{m_1^2 + m_2^2 + 2m_1m_2\gamma_1\gamma_2(1 - \frac{u_1u_2}{c^2})}, \quad u_3 = \frac{m_1\gamma_1u_1 + m_2\gamma_2u_2}{m_1\gamma_1 + m_2\gamma_2}$$

$$\text{where } \gamma_i = \left(1 - \frac{u_i^2}{c^2}\right)^{-1/2}. \quad [6]$$

Question 3.

- (a) Establish the transformation formulae for energy and momentum. [6]
 - (b) What do you understand by the relativistic Hamiltonian? [6]
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Question 4.

- (a) Obtain the transformation formulae for the densities of electric charge and electric current. [6]
 - (b) Prove the Lorentz invariance of Maxwell's equations without using tensors. [6]
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Question 5.

- (a) Obtain the transformation equations for $\vec{A}$ and ϕ , where $\vec{A}$ is the electromagnetic vector potential and ϕ is the scalar potential. [6]
 - (b) Find all components of the anti-symmetric tensor $F^{\mu\nu}$ in four dimensions, in terms of $\vec{E}$ and $\vec{H}$. [6]
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Question 6. Obtain the expression for the electromagnetic energy-momentum tensor in Minkowski space. [12]

Question 7.

- (a) Write a short note on the equivalence of mass and energy ($E = mc^2$). [6]
 - (b) Using the Lorentz condition, show that $\square^2\phi = 4\pi\rho$ and $\square^2\vec{A} = \frac{4\pi}{c}\vec{J}$. [6]
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